

Which Sources of Pollution Harm Coral Reefs in the Caribbean?

Coral reef ecosystems are increasingly threatened by climate change and destructive human activities. Understanding which ones actively hurt Caribbean coral reefs could help ecosystem managers, communities & governments take effective action to mitigate human impacts. Here, I ask:



To what extent does pollution correlate with coral bleaching and coral disease incidence in the Caribbean?



Figure 1: Mean Coral Bleaching by Pollution Level

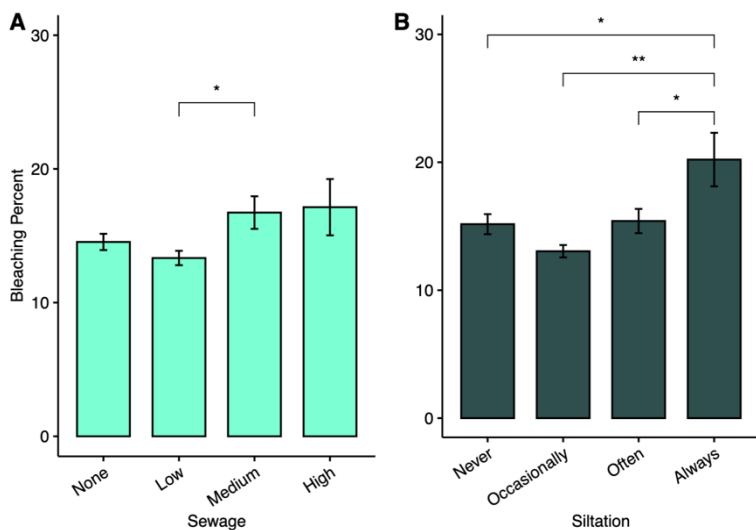
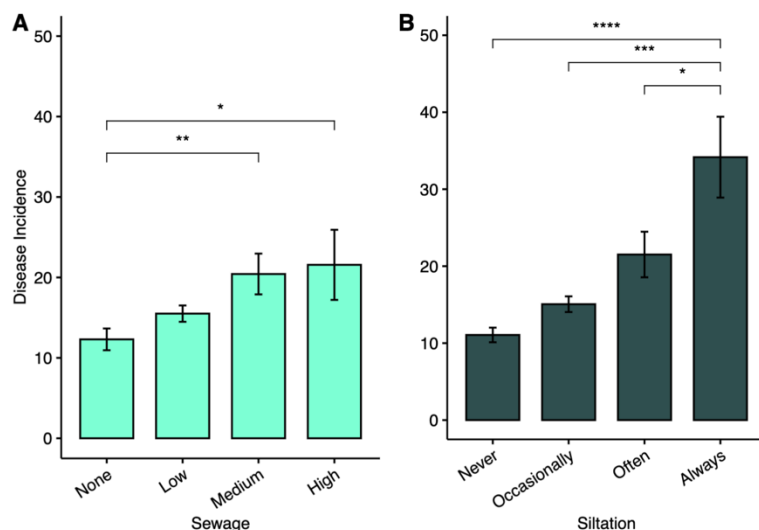


Figure 2: Mean Coral Disease Incidence by Pollution Level



Analysis of variance revealed a statistically significant difference in coral bleaching by level of sewage pollution ($F(3, 2837) = 3.43, p = 0.016$) and siltation ($F(3, 2856) = 7.34, p < 0.001$). Tukey's post hoc tests revealed that coral reefs with medium sewage pollution had significantly more coral bleaching than reefs with low sewage pollution ($p = 0.036$). Reefs that always experienced siltation had significantly greater bleaching than reefs that experienced siltation often ($p = 0.047$), occasionally ($p < 0.001$), or never ($p = 0.021$). Error bars indicate standard error. * indicates statistical significance at $p < 0.05$ and ** at $p < 0.01$.

Analysis of variance showed a significant difference in coral disease incidence by level of sewage ($F(3, 781) = 4.64, p = 0.0032$) and siltation ($F(3, 802) = 19.96, p < 0.0001$). Tukey's post hoc tests showed that reefs with no sewage pollution had significantly less disease incidence than reefs with medium sewage ($p = 0.0085$) and high sewage ($p = 0.044$). Reefs with constant siltation had significantly higher levels of disease incidence than reefs that often had siltation ($p = 0.0057$), occasionally ($p < 0.0001$), and never ($p < 0.0001$). Error bars indicate standard error. * indicates statistical significance at $p < 0.05$, ** at $p < 0.01$, and *** at $p < 0.001$.

In the Caribbean, coral reefs suffering from pollution frequently have greater levels of bleaching and disease, key indicators of coral health. **In particular, persistent siltation consistently correlates with significantly higher levels of bleaching and disease.** Further research should examine whether reducing siltation burden in these areas could help stressed coral reefs recover.

Data Source: Reef Check Foundation, Reef Check Global Reef Dataset. data.reefcheck.org. Accessed March 2, 2026.

Thank you to the Reef Check Foundation, its chapters, and its volunteers, who have spent countless hours collecting this data. This project would not have been possible without their contribution.

